Functions & Transformations

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Domain and range

State the domain of $f(x) = \sqrt{x - 3}$.

2 [Foundational] • Target: Domain and range

State the range of $f(x) = x^2 + 2$.

3 [Foundational] • Target: Transformations: shifts

The graph of $y = x^2$ is shifted 3 units right and 2 units down. Write the new equation.

4 [Foundational] • Target: Transformations: shifts

The graph of $y = |x|$ is shifted 4 units left and 1 unit up. Write the new equation.

5 [Foundational] • Target: Transformations: shifts

Describe the transformation that takes $y = x^2$ to $y = (x + 2)^2 - 5$.

6 [Foundational] • Target: Transformations: reflections and stretches

The graph of $y = x^2$ is reflected over the x-axis and stretched vertically by a factor of 3. Write the new equation.

7 [Foundational] • Target: Composite functions

Using the same f and g, find $(g \circ f)(2)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Composite functions

Using the same f and g, find a general formula for $(f \circ g)(x)$.

9 [Intermediate] • Target: Inverse functions

Find the inverse of $f(x) = 3x - 2$.

10 [Intermediate] • Target: Inverse functions

Show that $f(x) = 2x + 6$ and $g(x) = \frac{x - 6}{2}$ are inverses by computing $f(g(x))$.

11 [Intermediate] • Target: Piecewise functions

A function is defined by $f(x) = x + 2$ for $x < 1$, and $f(x) = 3x - 1$ for $x \geq 1$. Find $f(-2)$.

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12 [Intermediate] • Target: Piecewise functions

Using the same piecewise function, find $f(1)$.

13 [Intermediate] • Target: Even and odd functions

Determine whether $f(x) = x^2 - 3$ is even, odd, or neither.

14 [Intermediate] • Target: Increasing, decreasing, and average rate of change

Find the average rate of change of $f(x) = x^2$ from $x = 1$ to $x = 4$.

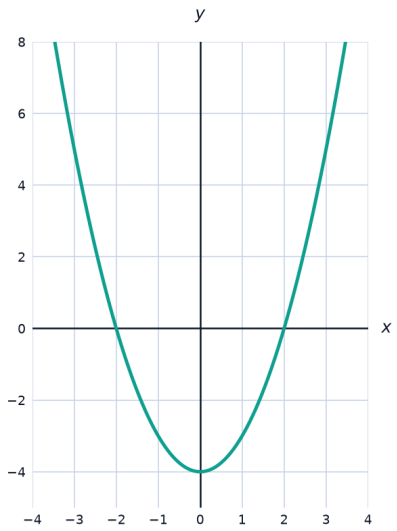
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Increasing, decreasing, and average rate of change • Cross-Lesson: 3.01 Sequences & Series

State the interval on which $f(x) = (x - 2)^2$ is decreasing.

16 [Challenging] • Target: Reading a function graph • Cross-Lesson: 4.01 Trigonometry

The graph of $y = f(x) = x^2 - 4$ is shown.



a State the zeros of f.

b State the domain and range of f.

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17 [Challenging] • Target: Word problems: functions in context • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

This question has two parts. A company's daily profit (in dollars) from selling x units is modeled by $P(x) = -2x^2 + 80x - 150$.

a Evaluate $P(10)$ to find the profit from selling 10 units.

b Find the number of units x that maximizes profit (using the vertex of the parabola), and state the maximum profit.

18 [Challenging] • Target: Word problems: functions in context • Cross-Lesson: 6.01 Limits & Derivatives

This question has two parts. The Fahrenheit conversion of a Celsius temperature is $F(C) = \frac{9}{5}C + 32$. Meanwhile, the temperature in a city t hours after 6 a.m. is modeled by $C(t) = -0.5t^2 + 6t + 10$ (in degrees Celsius), for $0 \le t \le 12$.

a Find $C(4)$, the Celsius temperature 4 hours after 6 a.m.

b Find $(F \circ C)(4)$, the Fahrenheit temperature at that time.

19 [Challenging] • Target: MCQ — Piecewise functions • Cross-Lesson: 7.01 Integration & Applications

Find the domain of the function $h(x) = \frac{1}{\sqrt{x-4}}$.

20 [Challenging] • Target: MCQ — Even and odd functions • Cross-Lesson: 8.01 Exponentials & Logarithms

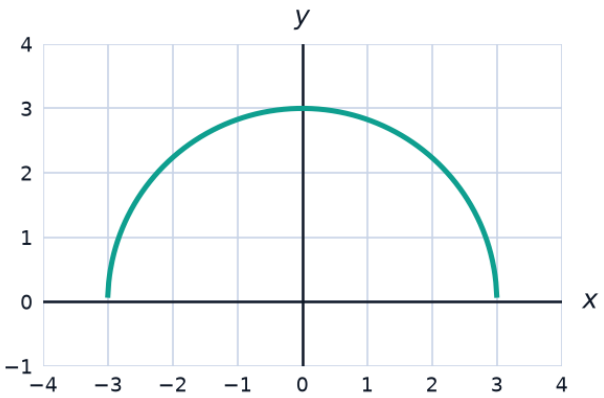
Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{2x-5}{3}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: MCQ — Domain and range

The graph shows $f(x)$ defined only where plotted. State its domain.



a. $[-3, 3]$

b. All real numbers

c. $[-4, 4]$

d. $[0, 3]$

22 [Foundational] • Target: MCQ — Transformations: shifts

The graph of $y = |x|$ is shifted 2 units left and 4 units up. Write the new equation.

a. $y = |x + 4| + 2$

b. $y = |x + 2| + 4$

c. $y = |x + 2| - 4$

d. $y = |x - 2| + 4$

23 [Foundational] • Target: MCQ — Transformations: shifts

The graph of $y = \sqrt{x}$ is shifted 6 units right. Write the new equation.

a. $y = \sqrt{x + 6}$

b. $y = \sqrt{x} - 6$

c. $y = \sqrt{x} + 6$

d. $y = \sqrt{x - 6}$

24 [Foundational] • Target: MCQ — Transformations: reflections and stretches

The graph of $y = \sqrt{x}$ is reflected over the y-axis. Write the new equation.

a. $y = -\sqrt{-x}$

b. $y = -\sqrt{x}$

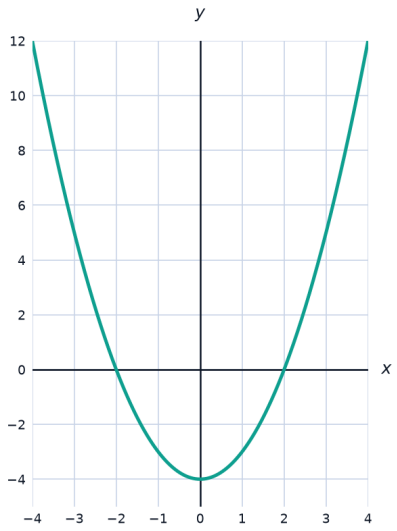
c. $y = \sqrt{-x}$

d. $y = \sqrt{x}$

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25 [Foundational] • Target: MCQ — Even and odd functions

The graph shown is symmetric about the y-axis. Is the function even, odd, or neither?



a. Odd

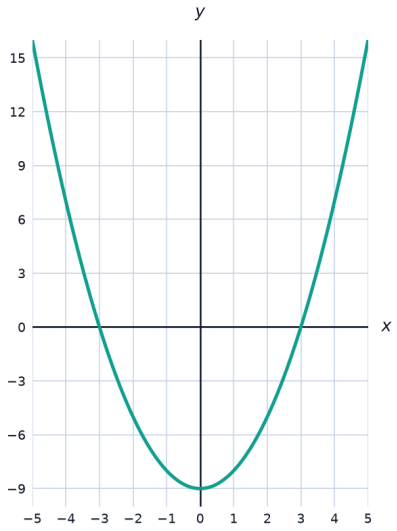
b. Neither

c. Cannot be determined

d. Even

26 [Foundational] • Target: MCQ — Reading a function graph

The graph shows $y = f(x) = x^2 - 9$. Find $f(3)$ by reading the graph.



a. -9

b. 6

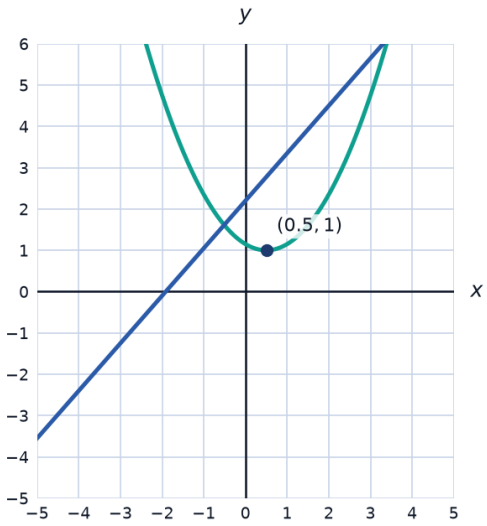
c. 9

d. 0

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27 [Foundational] • Target: MCQ — Reading a function graph

Using the same graph, find the values of x where $f(x) = 0$ (the x-intercepts).



a. $x = 0$ and $x = 9$

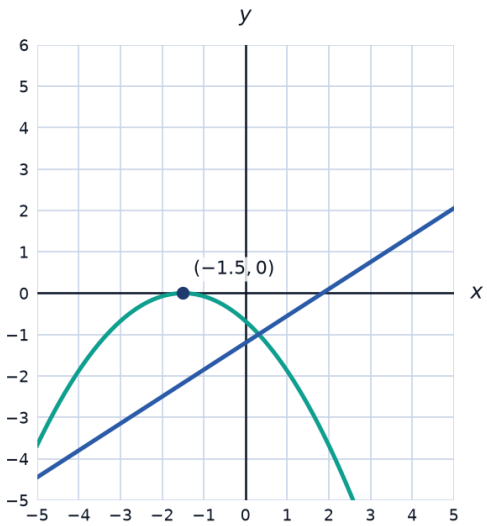
b. $x = -9$ and $x = 9$

c. $x = 3$ only

d. $x = -3$ and $x = 3$

28 [Foundational] • Target: MCQ — Reading a function graph

Using the same graph, find the y-intercept.



a. -3

b. 9

c. 0

d. -9

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: MCQ — Word problems: functions in context**

A company's daily profit (in dollars) from selling x units is modeled by $P(x) = -2x^2 + 80x - 300$. Find the profit when 15 units are sold.

a. 450 dollars

b. 150 dollars

c. 600 dollars

d. 900 dollars

30 **[Intermediate]** • **Target: MCQ — Word problems: functions in context**

The Fahrenheit conversion of a Celsius temperature is $F(C) = \frac{9C}{5} + 32$. Find $F(20)$.

a. 72°F

b. 68°F

c. 36°F

d. 52°F

31 **[Intermediate]** • **Target: MCQ — Word problems: functions in context**

A ball's height (in meters) t seconds after being thrown is $h(t) = -5t^2 + 20t + 2$. Find the height at $t = 2$ seconds.

a. 2

b. 18

c. 22

d. 42

32 **[Intermediate]** • **Target: Word problems: functions in context**

The graph of $y = x^2$ is shifted 1 unit left and 4 units down. Write the new equation.

a. $y = (x - 1)^2 + 4$

b. $y = (x + 1)^2 + 4$

c. $y = (x + 1)^2 - 4$

d. $y = (x - 1)^2 - 4$

33 **[Intermediate]** • **Target: MCQ — Evaluating functions**

The graph of $y = |x|$ is shifted 7 units left and 3 units down. Write the new equation.

a. $y = |x + 7| + 3$

b. $y = |x - 7| - 3$

c. $y = |x + 7| - 3$

d. $y = |x - 7| + 3$

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34 [Intermediate] • Target: MCQ — Domain and range

[redacted] $f(x) = x^2 - 3x$ [redacted] [redacted] $[1, 4]$[redacted] Average rate of change of $x^2 - 3x$ on $[1, 4]$?

a. 2

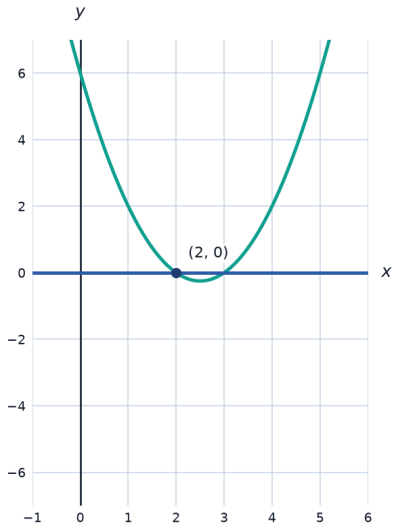
b. 3

c. 4

d. 5

35 [Intermediate] • Target: MCQ — Transformations: shifts

[redacted] $f(x) = x^2 - 5x + k$. [redacted] [redacted] [redacted] [redacted] x [redacted] $x = 2$[redacted] [redacted] [redacted] k[redacted] The graph below represents the function $f(x) = x^2 - 5x + k$. If the curve intersects the x-axis at $x = 2$, what is the value of k?



a. –6

b. 6

c. –4

d. 4

36 [Intermediate] • Target: MCQ — Transformations: reflections and stretches

[redacted] $f(x) = \sqrt[3]{x - 5}$[redacted] Domain of cube root function $\sqrt[3]{x - 5}$?

a. $\mathbb{R}$

b. $[5, \infty)$

c. $(5, \infty)$

d. $\mathbb{R} \setminus \{5\}$

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37 [Intermediate] • Target: MCQ — Composite functions

[redacted] Which is an odd function symmetric about origin?

a. $f(x) = x^3 - x$

b. $f(x) = x^2 - 1$

c. $f(x) = x^4 + 3$

d. $f(x) = |x|$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Inverse functions • Cross-Lesson: 6.01 Limits & Derivatives

[redacted] y[redacted] Which is an even function symmetric about y-axis?

a. $f(x) = x^2 + 4$

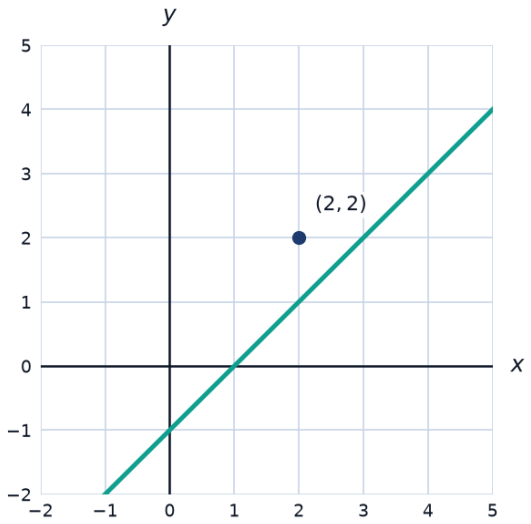
b. $f(x) = x^3$

c. $f(x) = x^3 + x$

d. $f(x) = 2x + 1$

39 [Challenging] • Target: MCQ — Piecewise functions • Cross-Lesson: 7.01 Integration & Applications

[redacted] $f(x)$ [redacted]. [redacted] $\lim_{x \to 2} f(x)$[redacted] The graph of the function $f(x)$ is shown below. What is the value of $\lim_{x \to 2} f(x)$?



a. 0

b. 1

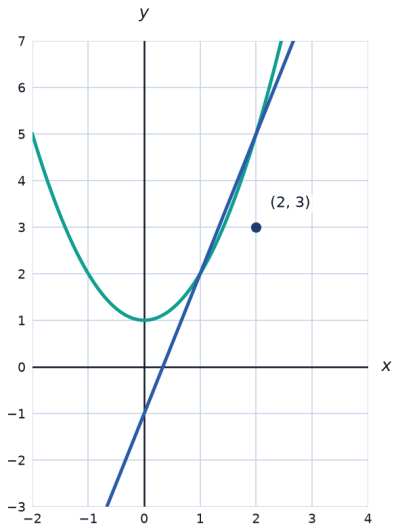
c. 2

d. 4

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40 [Challenging] • Target: MCQ — Even and odd functions • Cross-Lesson: 8.01 Exponentials & Logarithms

[redacted] The graph of the function $f(x)$ and its tangent line at $x = 2$ is shown below. If the equation of the tangent line is $y = 3x - 1$, what is the value of $f'(2)$?



a. 1

b. 2

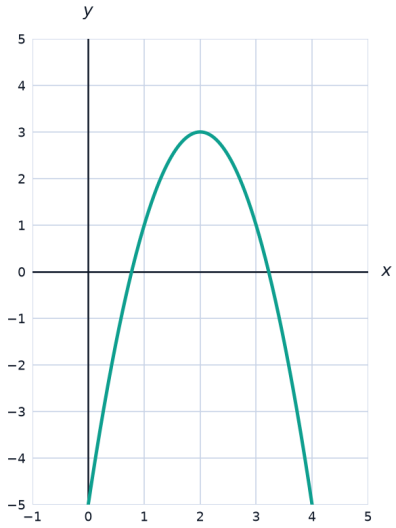
c. 3

d. -1

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41 [Challenging] • Target: MCQ — Increasing, decreasing, and average rate of change • Cross-Lesson: 9.01 Statistics & Probability

[redacted] $f(x) = -2x^2 + 8x - 5$ [redacted]. [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] The function $f(x) = -2x^2 + 8x - 5$ represents the trajectory of a projectile. What is the maximum height the projectile reaches?



a. 4 [redacted] 4 units

b. 3 [redacted] 3 units

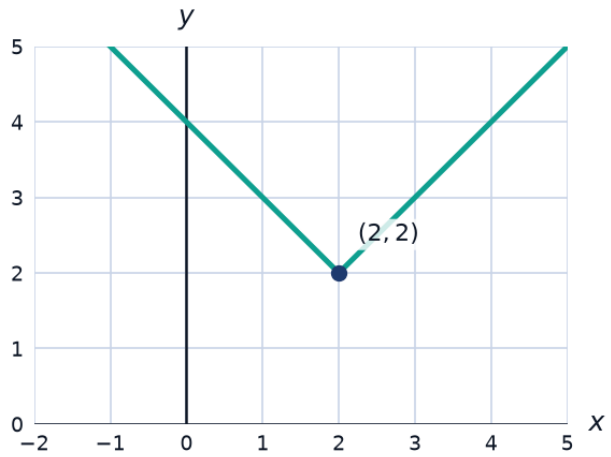
c. 5 [redacted] 5 units

d. 11 [redacted] 11 units

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42 [Challenging] • Target: MCQ — Operations on functions • Cross-Lesson: 1.01 Algebra & Equations

[redacted] $f(x)$ [redacted] $x = 2$. [redacted] $f(x)$ [redacted] x [redacted] 2[redacted] The graph of the function $f(x)$ is shown. What is the limit of $f(x)$ as x approaches 2?



a. -1

b. 2

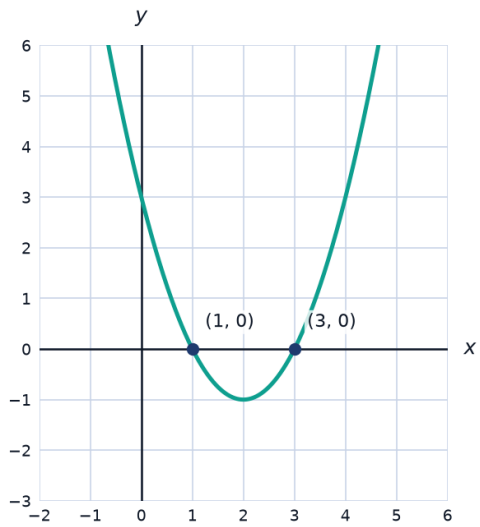
c. 0

d. 3

43 [Challenging] • Target: MCQ — Reading a function graph • Cross-Lesson: 3.01 Sequences & Series

[redacted] $f(x) = x^2 - 4x + 3$[redacted]

Given the function $f(x) = x^2 - 4x + 3$, what are its x-intercepts?



a. $(1, 0)$ ■ $(3, 0)$

b. $(-1, 0)$ ■ $(-3, 0)$

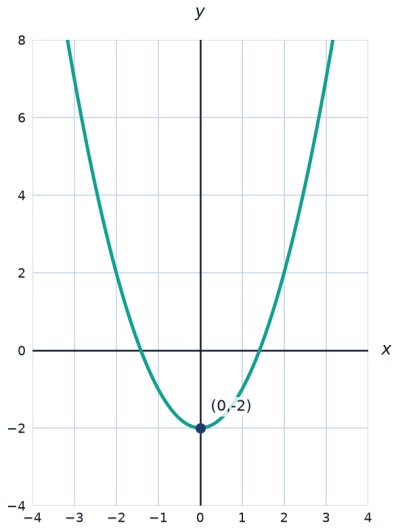
c. $(1, 0)$ ■ $(-3, 0)$

d. $(-1, 0)$ ■ $(3, 0)$

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44 [Challenging] • Target: MCQ — Word problems: functions in context • Cross-Lesson: 4.01 Trigonometry

■■■ ■■■■ $f(x) = x^2 - 2$ ■ $g(x) = \sqrt{x}$ ■ ■■■ ■■ ■■■■ $(f \circ g)(x)$ ■■■■■■■■ Formula and domain of $(f \circ g)(x)$?



a. $x - 2$ ■■■■■■■ $[0, \infty)$

b. $x - 2$ ■■■■■■■ $\mathbb{R}$

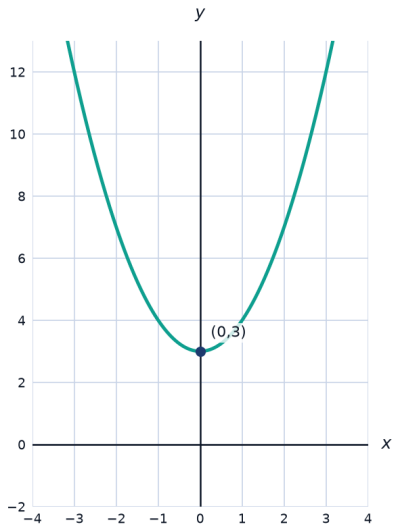
c. $\sqrt{x^2 - 2}$ ■■■■■■■ $[0, \infty)$

d. $x^2 - 2$ ■■■■■■■ $\mathbb{R}$

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45 [Challenging] • Target: Evaluating functions • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

■■ ■■■ ■■■■■■ $f(x) = x^2 + 3$■ Range of $f(x) = x^2 + 3$?



a. $[3, \infty)$

b. $(3, \infty)$

c. $[0, \infty)$

d. $\mathbb{R}$

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